Experiment 1: Study of Electronics Components

Resistors

A resistor is a component of an electrical circuit that resists the flow of electrical current. A resistor has two terminals across which electricity must pass, and is designed to drop the voltage of the current as it flows from one terminal to the next. A resistor is primarily used to create and maintain a known safe current within an electrical component.



Physical_Structure



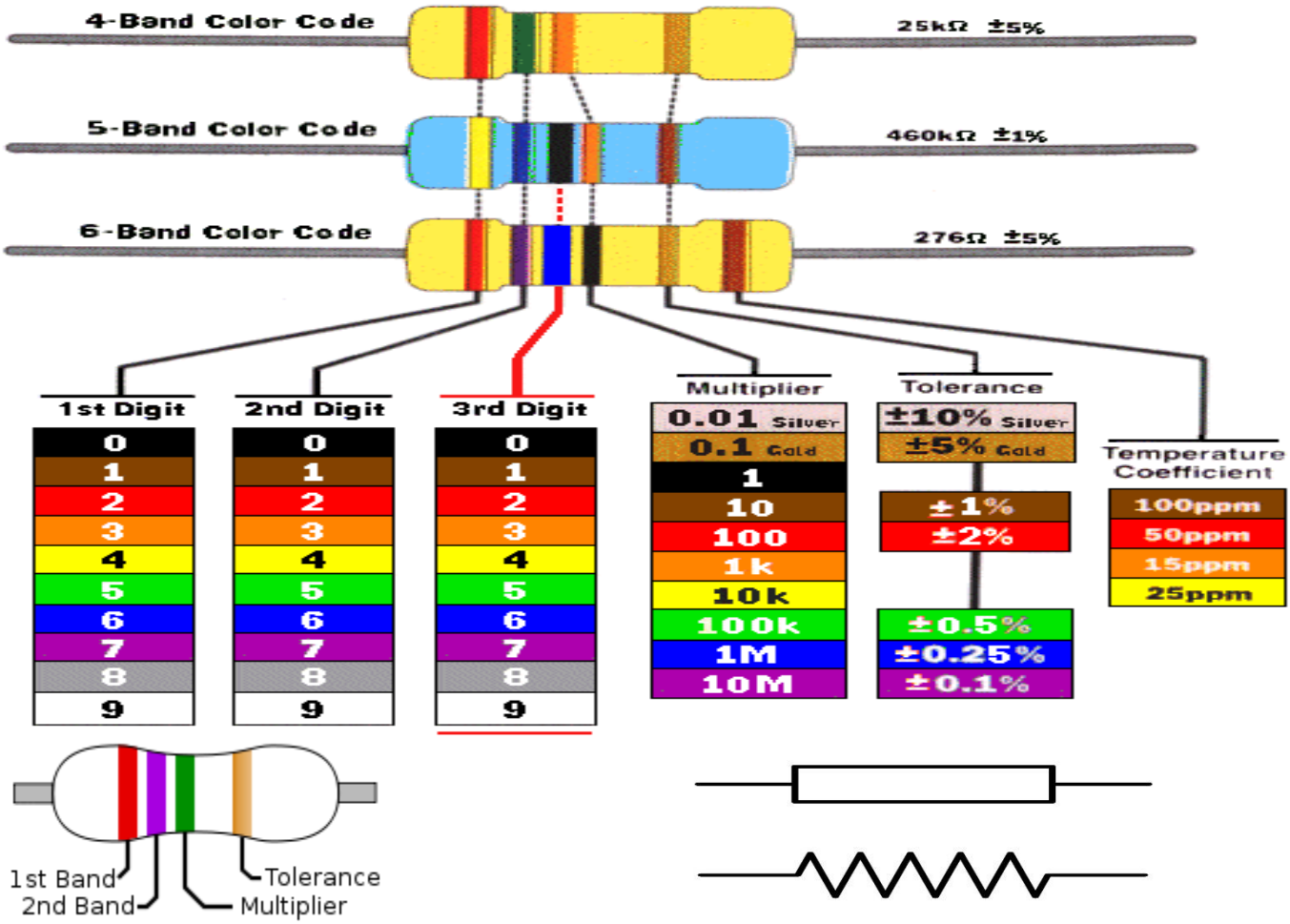
Symbol

A. **FIXED VALUE RESISTOR (Resistor Color Code Chart)**

Band Color Options	Band 1 position No	Band 2 position No	Band 3 Multiplier Value	Band 4 Value Tolerance
Black	0	0	×10°	---
Brown	1	1	×10¹	± 1%
Red	2	2	×10²	± 2%
Orange	3	3	×10³	---
Yellow	4	4	×10⁴	----
Green	5	5	×10⁵	± 0.5%
Blue	6	6	×10⁶	± 0.25%
Violet	7	7	--	± 0.1%
Gray	8	8	--	± 0.05%
White	9	9	--	----
None	--	--	--	± 20%
Silver	--	--	× 0.01	± 10%
Gold	--	--	× 0.1	± 5%

Example: - (Measurement process) -Most resistors have 4 bands:

- The **first band** gives the **first digit**.
- The **second band** gives the **second digit**.
- The **third band** indicates the **number of zeros**.
- The **fourth band** is used to show the tolerance (precision) of the resistor



Calculation	First Band	Second Band	Third Band	Fourth Band	Calculated Value
Color Name	Red	Violet	Green	Silver	2700000Ω ±10%
Position Value	2	7	10^5	±10%	2.7MΩ ±10%

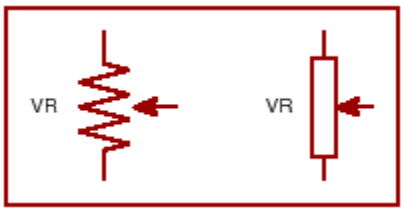
Resistor	Calculation	First Band	Second Band	Third Band	Fourth Band	Calculated Value	Measurement Value by Digital Multimeter
Sample #1	Color Name: - Position Value: - Calculated Value: -						
Sample #2	Color Name: - Position Value: - Calculated Value: -						
Sample #3	Color Name: - Position Value: - Calculated Value: -						

C. Function: -

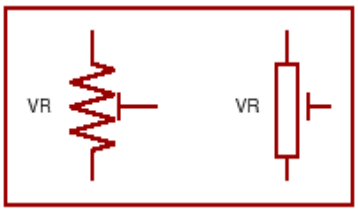
D. Specification: - 1. Power Rating (W) 2.Resistive Value (Ω Value)

VARIABLE RESISTOR

1. Potentiometer Symbol



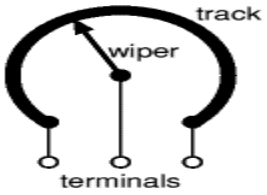
2. Preset Symbol



Physical Structure:







	Standard Value (Written on the Device Surface)	The measured value by Multimeter (between two end terminals)
Potentiometer		
Preset		

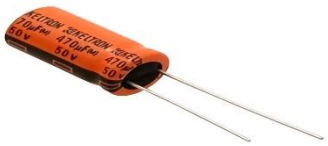
*** test the variable resister value (any one end terminal and middle terminal) by multimeter. It will show the value from 0 Ω to the maximum value which is defined on the device surface.

CAPACITORS

Polarized (Electrolytic Capacitors, Value Above 1uF)

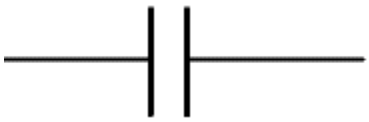


Circuit Symbol



Physical Structure

Unpolarized (Non- Electrolytic) Small Value below 1uF



Circuit Symbol



Physical structure

Capacitor	Measure the standard value /Calculate the value (written on the device surface)	The measured Value by LCR meter
Electrolytic Capacitor		
Non-Electrolytic Capacitor	Smaple-1	

	Sample-2 Sample-3	

Capacitor Number Code (for non-electrolytic)



A number code is often used on small capacitors where printing is difficult:

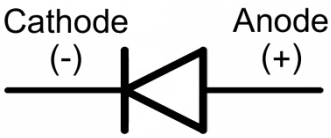
- the 1st number is the 1st digit,
- the 2nd number is the 2nd digit,
- the 3rd number is the number of zeros to give the capacitance in pF.
- Ignore any letters - they just indicate tolerance and voltage rating.

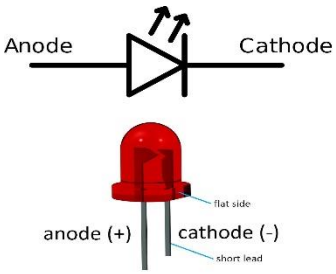
For example: **102** means 1000pF = 1nF (*not 102pF!)*
If there is no third digit (only two digit), then specified number is the capacitive value in pF.
Three prefixes (multipliers) are used, µ (micro), n (nano) and p (pico):

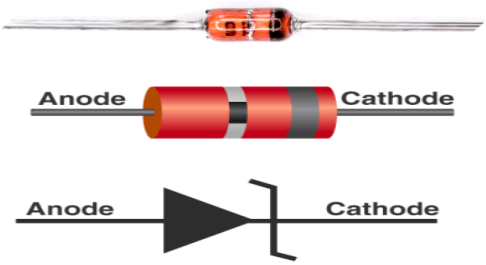
- µ means 10^{-6} (millionth), so 1000000µF = 1F
- n means 10^{-9} (thousand-millionth), so 1000nF = 1µF
- p means 10^{-12} (million-millionth), so 1000pF = 1nF

Specification: - **1. Voltage Rating: -** **2. Capacitor's Rating: -**

DIODE





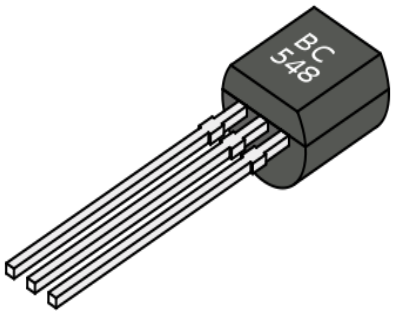


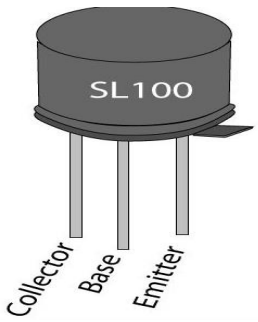
(Forward and Reverse voltage drop will be measured by multimeter in diode testing mode)

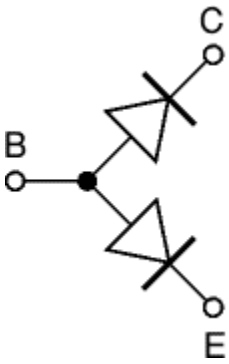
	Symbol	Device Number	Forward Voltage Drop	Reverse Voltage Drop	Determine the anode and cathode terminal.
Silicon Diode					
Zener Diode					
LED (Specify the Color)					

**** Diode Specification and Parameters will be noted from the Datasheet (according to the device no. which is given on the device surface).

Bipolar Junction Transistor (BJT)









The Leads are labeled **base (B), Collector(C), and Emitter(E).**

Transistor Testing with a Digital multimeter

Same as diode test. For a Transistor is nothing but two diodes connected back-to-back.
Note down the transistor number which is specified on the device surface.

******Test each pair of leads both ways (six tests in total):**

- The **base-emitter (BE)** junction should behave like a diode and conduct one way only.
- The **base-collector (BC)** junction should behave like a diode and conduct one way only.
- The **collector-emitter (CE)** should not conduct either way.
- You have to note how much forward bias voltage from these six tests.

Then conclude that the transistor is PNP or NPN

	Type of transistor (PNP or NPN)	Symbol	Voltage C to B terminal	Voltage B to C terminal	Voltage C to E terminal	Voltage E to C terminal	Voltage B to E terminal	Voltage E to B terminal
BC548								
SL100								

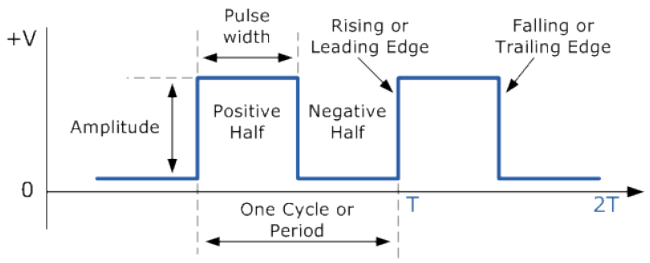
**** Only two voltages will show forward bias diode voltage.**

Experiment 2: Instrument Study

Operation of Function Generator, DSO, Power supply and Multimeter

A. Display the internally available waveform from DSO. And measure its parameters

1. Time period(T)
2. Amplitude (Vm) or V_{max}
3. Voltage peak-peak (Vpp)
4. Frequency(F)
5. Rise time (Tr) and fall time (Tf)
6. Trace or draw the waveform

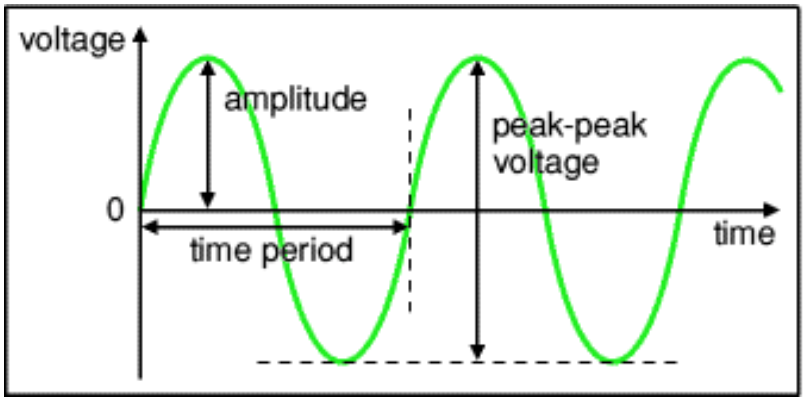


This is a DSO inbuilt waveform. Measure the parameters and trace the waveform.
** change the Volts/division and time/division knob and observe the change in display.

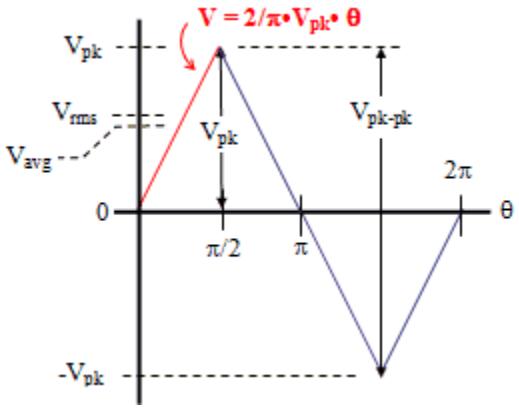
B. Connect Function generator output probe with DSO input probe (ch-1 or ch-2)

All the parameters set in Function Generators		**Display and measure by DSO and draw any one of each of the waveforms.**
Type 1.Sinewave	Frequency→Vpp 70Hz→.1V, 650Hz→1.2V, 1.2KHz→3.2V, 15KHz→5V 60KHz→10V	Parameters 1. Time period(T) 2. Voltage peak-peak (Vpp) 3. Frequency(F) 4.Rise time (Tr) and fall time (Tf)
2. Square wave	same	
3. Triangular wane	same	

Sine wave (Draw at least one sine wave with dimension)



Triangular Wave (Draw one triangular wave with dimension)



C. Power Supply: Operate a power supply to generate the following voltages. Measure these DC voltages using a digital multimeter. Voltage ranges: ±5vdc, ±12vdc, 1v,2v,10v,15volts

Experiment 3 V-I Characteristics of Semiconductor Diode (Si)

V-I Characteristics of Semiconductor Diode (Si)

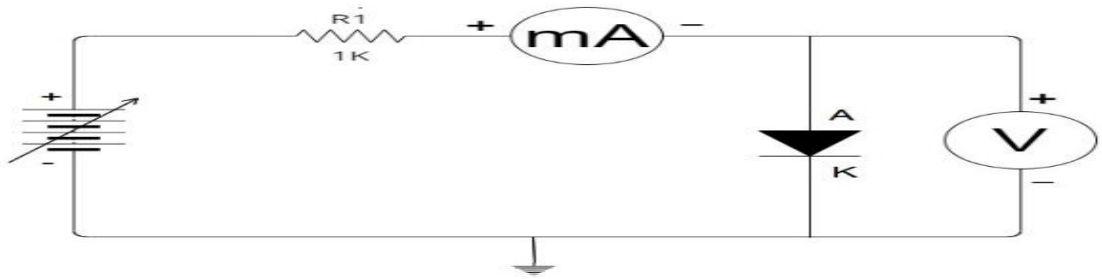
Objective: -

- Design the circuit diagram, Draw the characteristics curve & Determine the DC forward resistance

Equipment's required: -

- Voltmeter or Digital Multimeter, Ammeter, Bread Board, Diode (IN4007), Resistors (1k Ohm) & 0 - 30 V dc power supply

Forward Bias Circuit Diagram



Theory: -

In electronics, a **diode** is a two-terminal electric component that conducts electric current in only one direction. The term usually refers to a **semiconductor diode**, the most common type today, which is a crystal of semiconductor connected to two electrical terminals. Today most diodes are made of silicon, but other semiconductors such as germanium are sometimes used.

The Shockley ideal diode equation or the diode law: -

$$I = I_S \left(e^{V_D/(nV_T)} - 1\right),$$

Where

I is the diode current,

$\boldsymbol{I_S}$ is the reverse bias saturation current,

$\boldsymbol{V_D}$ is the voltage across the diode,

$\boldsymbol{V_T}$ is the thermal voltage, and

n is the emission coefficient, also known as the ideality factor. The emission coefficient varies from about 1 to 2 depending on the fabrication process and in many cases is assumed to be approximately equal to 1.

$$V_T = \frac{kT}{q},$$

where ***k*** is the Boltzmann constant, ***T*** is the absolute temperature of the **P-N** junction, and ***q*** is the magnitude of charge on an electron. $\boldsymbol{V_T}$ is approximately 25.85 mV at 300 K.

DC forward resistance:

It is the opposition by diode to the DC. It is measured by the ratio of DC voltages across the diode to the resulting DC current through it.

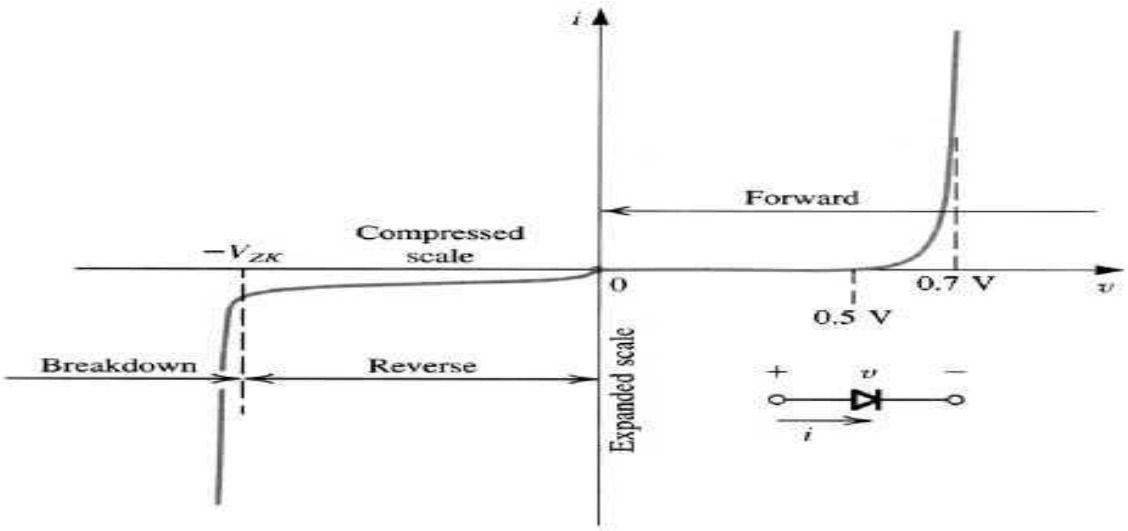
Procedure: -

1. Connect the circuit as shown in the figure.
2. Increase the supply voltage by 0.5v. Then measure current $\mathbf{I_D}$, $\mathbf{V_d}$ and record the results in Table
3. Repeat step 2 till DC supply of 5v.
4. Plot on a graph paper $\mathbf{I_D}$ versus $\mathbf{V_D}$ for the silicon diode. Complete the curves by extending the lower region of each curve to the intersection of the axis at $\mathbf{I_D}$ = 0 mA and $\mathbf{V_D}$ = 0 V.

Result Table (Si)

V_{source} (up to 0.5 volts increase in steps of .1v to see a smooth curve)	$\mathbf{V_d}$	$\mathbf{I_d}$
0.5V		
1V		
1.5V		
2V		
2.5V		
3V		
3.5		
4V		
4.5V		
5V		

Expected I-V characteristics:



The diode i–v relationship with some scales expanded and others compressed in order to reveal details.

V-I Characteristics of Zener Diode

Objective: -

- Design the circuit diagram.
- Take the reading and draw the characteristics curve. Determine the DC forward and Reverse resistance, Determine the Zener Voltage (Vz)

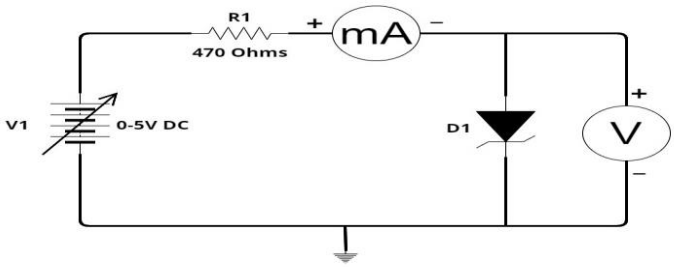
Equipment's required: -

- Voltmeter or Digital multimeter, Amp meter, Bread Board, Zener Diodes, Resistors (1k Ohm), 0 - 30 V dc power supply

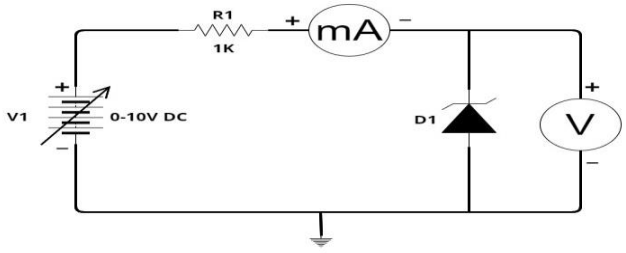
Theory: -

A **Zener diode** is a type of diode that permits current not only in the forward direction like a normal diode, but also in the reverse direction if the voltage is larger than the breakdown voltage known as "Zener knee voltage" or "Zener voltage". The forward characteristic of the Zener diode is same as that of a *PN* junction diode i.e. as the applied potential increases the current increases exponentially. Applying a negative potential to the anode and positive potential to the cathode reverse biases the Zener diode. As the reverse bias increases, the current increases rapidly in a direction opposite to that of the positive voltage region. Thus, under reverse bias condition breakdown occurs.

Forward Bias Circuit Diagram



Reverse Bias Circuit Diagram



Procedure: -

Forward-Bias: -

1. Construct the circuit as shown in figure for forward biased Zener diode.
2. Take the reading from the voltmeter and the amp meter from the output side and fill the table by increasing the input DC voltage.
3. Plot $\mathbf{I_d}$ vs $\mathbf{V_d}$ in a graph to get the forward biased characteristics.

Reverse Bias: -

1. Construct the circuit as shown in figure for reverse biased Zener diode.

2. Take the reading from the voltmeter and the amp meter from the output side and fill the table by increasing the input DC voltage.
3. Plot $\mathbf{I_d}$ vs $\mathbf{V_d}$ in a graph to get the reverse biased characteristics.
4. Find the voltage at the point where $\mathbf{I_d}$ decreases sharply to get the Zener voltage.

Result Table (Forward Bias)

V source	Vd	Id
.5V		
1V		
1.5V		
2V		
2.5V		
3V		
3.5		
4V		
4.5V		
5V		

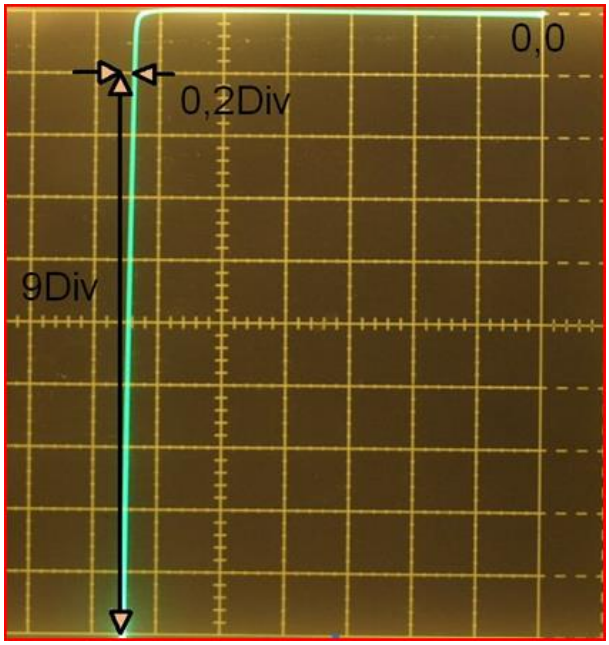
Reverse Biass

V source	V_d	I_d
1V		
2V		
3V		
4V		
5V		
6V		
7V		
8V		
9V		
10V		

Expected I-V characteristics (Forward Bias): -



Expected I-V Characteristics (Reverse bias): -



Experiment 4 Study of Half wave and Full wave Rectifier

Study the Half wave &Full wave Bridge rectifier

Objective: - Design the circuit diagram. Take the reading and draw the input & output waveform. Determine the output voltage with filter circuit across load (RL).

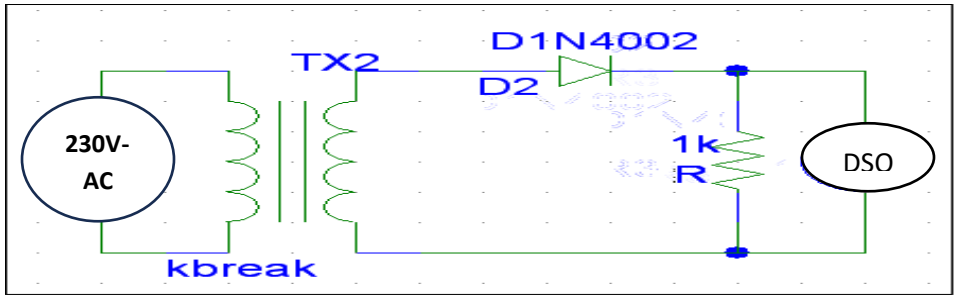
Equipment's required: -1) Digital Multimeter 2) DSO 3) Universal Work Station with 5v AC supply.

Components: - Diodes (IN4007) – 4 Nos, Resistors 1k-1no., Capacitor 100uF-1no.

Theory:-A **rectifier** is an electrical device that converts alternating current (AC) to direct current (DC).In half wave rectification, either the positive or negative half of the AC wave is passed, while the other half is blocked. A full-wave rectifier converts the whole of the input waveform to one of constant polarity (positive or negative) at its output.

Half-wave and full-wave rectifiers deliver a form of pulsating DC output. In order to produce steady DC from a rectified AC supply, a filter circuit is required. In its simplest form this is achieved by shunting the resistor with a capacitor. There will still remain an amount of AC ripple voltage where the ripple voltage is defined as the deviation of the load voltage from its average or dc value.

Circuit Diagram of Half wave rectifier



Procedure: -

Half Wave Rectification / Full wave rectification-

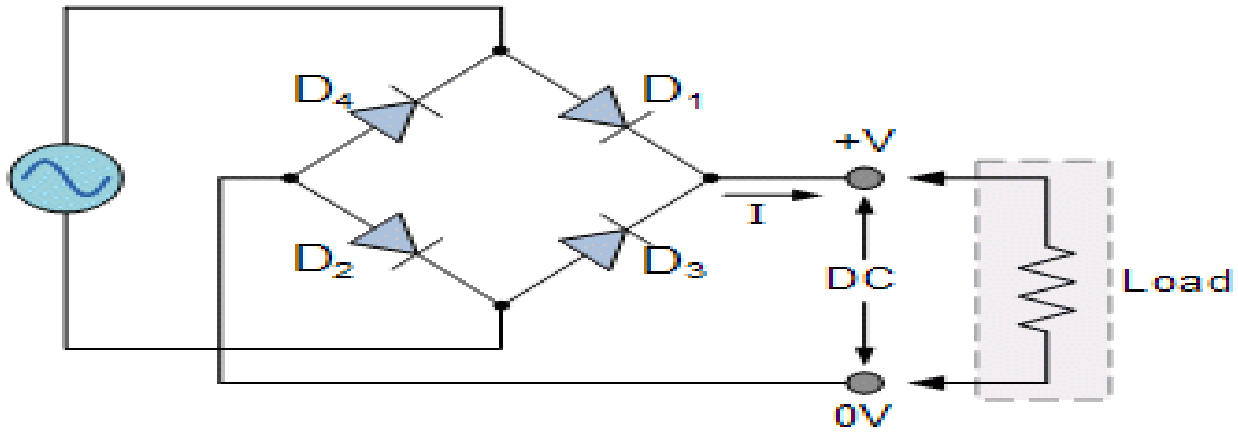
1. Construct the circuit as shown. Take the supply of 5 Vrms sinusoidal wave from the UWS with a frequency of 50 Hz. Put the oscilloscope probes at input and sketch the input waveform obtained.
2. Put the oscilloscope probes across the resistor and sketch the output waveform obtained. Measure and record the DC level of the output voltage.
3. Now connect the capacitor across the resistor and measure the filtered output voltage.
4. Also measure the Ripple Voltage.

ircuit Diagram of Full wave rectifier

Full-Wave-Rectification

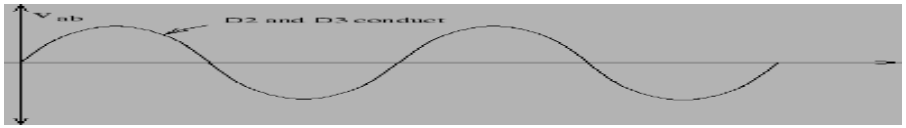
The diode D2 and D3 conduct in the positive half cycle while D1 and D4 in the negative half cycle. The output is the same as center tapped transformer based full wave rectifier. One important point about the full wave rectifier: the input & output do not share a common reference! For this reason, you cannot view both input & output simultaneously on an oscilloscope.

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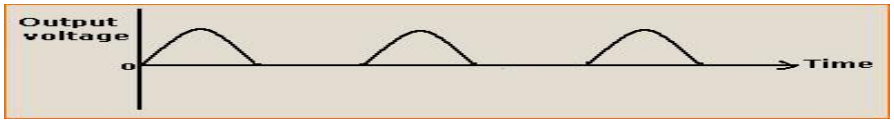


Type of rectifier	Input-AC voltage (Transformer secondary) V_{RMS} Vpk-pk, Vmax From DSO	Rectifier Output Vm (Vmax), Vrms, Vmean or Vdc from CRO (DC coupling mode of DSO)	Form factor Vrms/Vdc	Ripple factor $\sqrt{((Vrms/Vdc)^2-1)}$	Measure Output-dc voltage with filter Multimeter in dc mode.	Ripple output Voltage with filter Multimeter in ac mode. Vr(rms).	Theoretically calculate the dc value expected. Vdc= Vm -Δ/2 Where Δ =Vr(p-p) =2√3Vr(rms) (Vm is rectifier output from 3rd column)
Half Wave							
Full Wave							

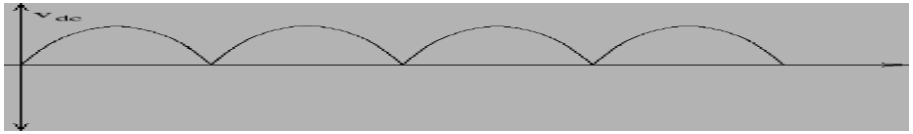
Expected input:



Expected Output of Half wave rectifier without filter: -



Expected Output of Full wave rectifier without filter:



Expected Output of Full wave rectifier with filter: -

B.2 RIPPLE VOLTAGE OF CAPACITOR FILTER

Assuming a triangular ripple waveform approximation as shown in Fig. B.1, we can write (see Fig. B.2)

$$V_{dc} = V_m - \frac{V_r(\text{p-p})}{2} \tag{B.3}$$

During capacitor discharge, the voltage change across C is

$$V_r(\text{p-p}) = \frac{I_{dc}T_2}{C} \tag{B.4}$$

From the triangular waveform in Fig. B.1,

$$V_r(\text{rms}) = \frac{V_r(\text{p-p})}{2\sqrt{3}} \tag{B.5}$$

(obtained by calculations not shown).

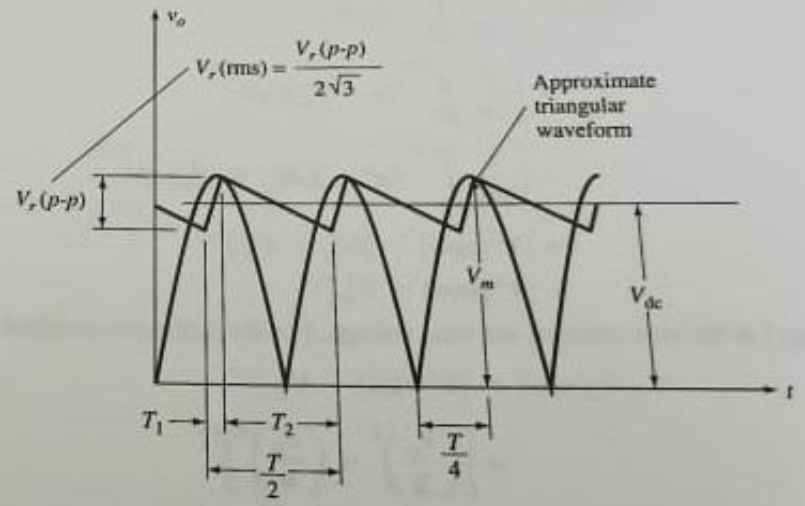


FIG. B.1

Approximate triangular ripple voltage for capacitor filter.

Experiment 5 Input and Output Characteristics of BJT

Input and output Characteristics of BJT

Objective: -

1. Design the circuit diagram.
2. Take the reading and draw the characteristics curve
3. Determine the input and output resistance.

Equipment's required: -

1)Voltmeter 2) Ammeter 3) Bread Board 4) 0 - 30 V dc power supply

Components required: -

1) Resistors 1k, 470K Ohm, 1M Ohm, Transistor (**BC547**)

Theory: -

A BJT is a three terminal electronic device consisting of either two *n-type* or one *p-type* layers of material or *two **p-type*** and one ***n-type*** layers of material. The former is called an ***NPN*** transistor and the latter is called ***PNP*** transistor. The three terminals are **Emitter (E), Collector(C) and Base (B).**

The Common Emitter (CE) configuration is most frequently encountered configuration. In this the emitter is common or reference to both the input and output terminals (in this case common to both the base and collector terminals). Two sets of characteristics are necessary to describe fully the behavior of the CE configuration: *input* and *output* characteristics.

The plot of base current I_B versus the voltage V_{BE} across its base and emitter for different values of voltage, V_{CE} is called the input characteristics.

The plot of collector current I_C versus the voltage V_{CE} across its collector and emitter for different values of base current I_B is called the output characteristics.

In the DC mode the levels of I_C and I_B are related by a quantity called beta (β) defined by

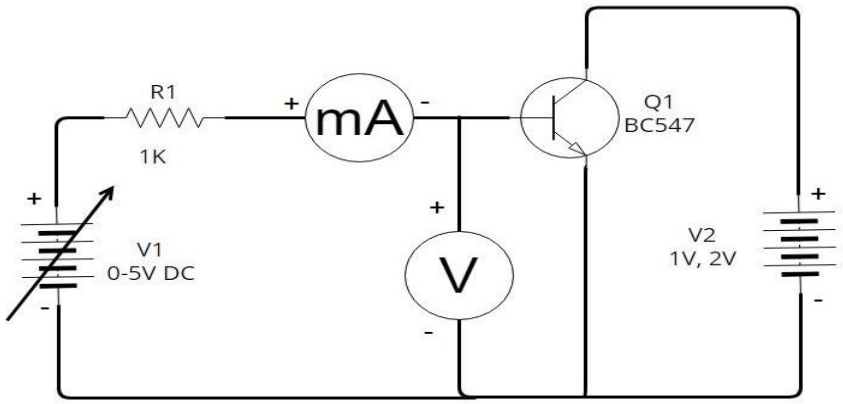
$$\beta_{dc} = \frac{I_C}{I_B}$$

For AC situations an ac beta is defined as

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}, \ \mathbf{V_{CE}= constant}$$

Both the beta values can be determined from the output characteristics curves.

Input characteristics Circuit Diagram



Procedure: -

Input characteristics

1. Set up the circuit as shown in the diagram.
2. V_{CE} (V2) is set at 1V.
3. Vary the input voltage V_1 in steps of 0.5V and note the corresponding I_B.
4. Repeat for V_{CE}(V2) =2V.
5. Plot the input characteristics curve with V_{BE} on x-axis and I_B on y-axis.

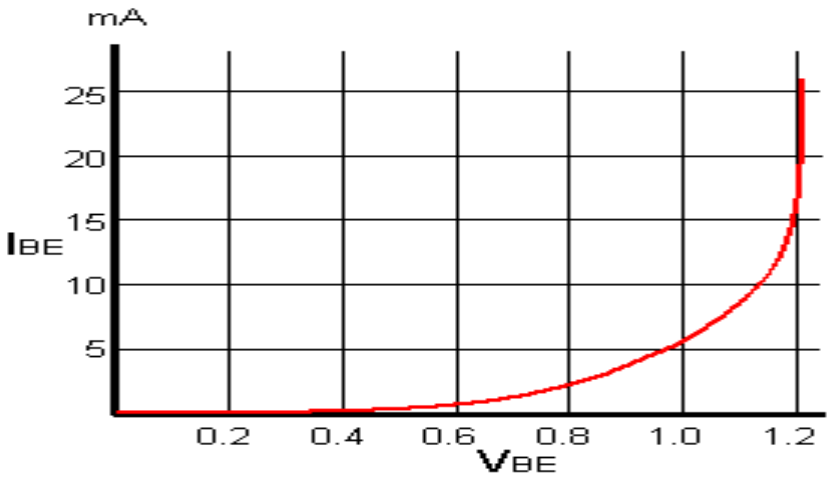
Result Table

Input Characteristics

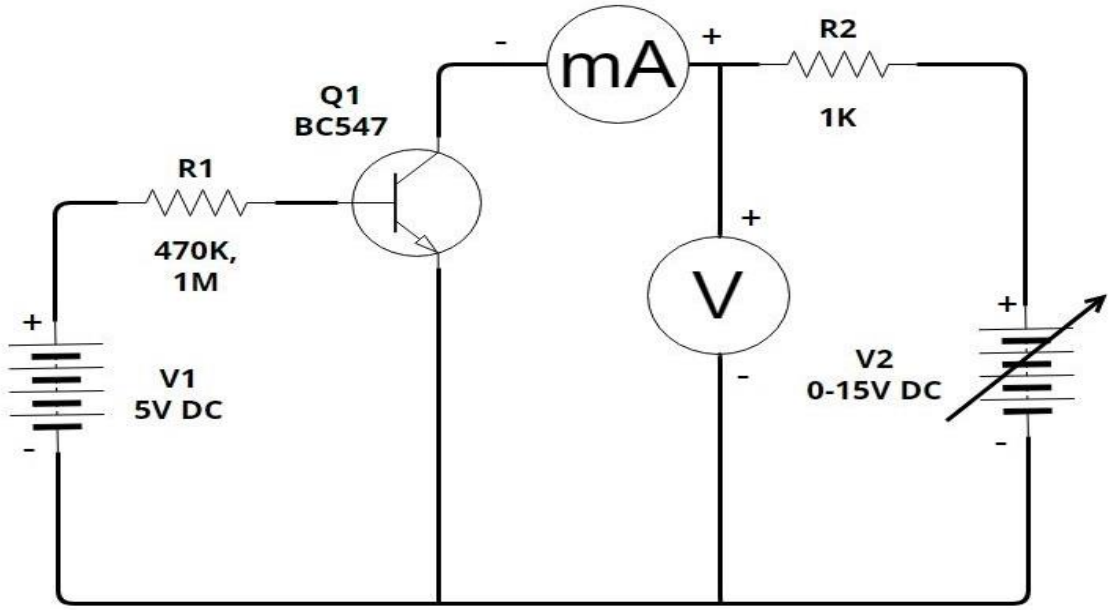
	V_{CE}=1V,2V	
V_{bb}(V1)	I_B	V_{be}
0V		
.5V		
1V		
1.5V		
2V		
2.5V		
3V		
3.5		
4V		
4.5V		
5V		

Expected Graphs

Input characteristics



Output Characteristics Circuit Diagram



Output characteristics

1. Set up the circuit as shown in the diagram.
2. R1 is taken 1M ohm so that I_B is set at 4.3uA.
3. Vary V_{CC} (V2) in steps of 1V and note the corresponding I_C & V_{CE}
4. Repeat for I_B = 10uA, by replacing R1 by 470 kilo Ohm.
5. Plot the output characteristics curve with V_{CE} on x-axis and I_C on y-axis.

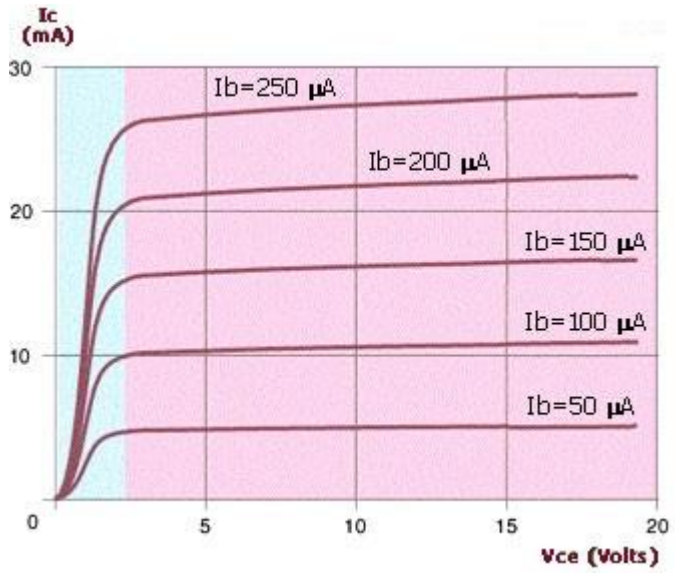
Output Characteristics

		I_B=10uA, 4.3uA
V_{CC}	V_{CE}	I_C

1V		
2V		
3V		
4V		
5V		
6V		
7V		
8V		
9V		
10V		

Expected Graphs

Output characteristics



Experiment 6 Design a RC Coupled CE Amplifier

Design a RC Coupled CE Transistor Amplifier

Objective: -

1. Connect the circuit diagram, Measure the DC Bias Conditions (V_{CE}, V_{BE}, I_C).
2. Measure the AC Gain, Signal Handling Capacity and the Bandwidth

Equipment's Required: -1) multimeter 2) Bread Board 3) 12V dc power supply 4) Function Generator 5) DSO

Components required: -1) Resistors (R1=68K, R2=18k, Rc=1.2K, Re=1K))2) Capacitors (Cin=C_{out}=1uF, C_e=33uF)

3) Transistor no- BC548 (NPN)

Theory: -An amplifier is a device that is used to amplify input voltage or current. The **RC coupled CE transistor amplifier** is a popular scheme of cascading two or more amplifiers to achieve the required levels of amplification. The inter-stage coupling is done by using a RC network, hence the name. The variation of gain with frequency of an amplifier is called the **frequency response characteristics** of the amplifier.

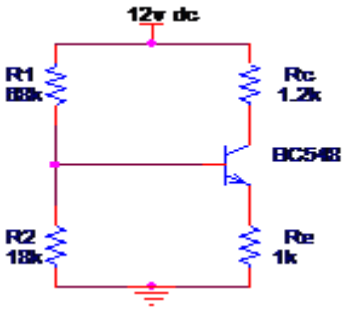
The **bandwidth** of the RC amplifier is the difference between **upper cut-off frequency** and **lower cut-off frequency**. It represents the range of frequencies that the amplifier is most effective in amplifying. The cut-off frequency points are determined from the frequency response graph, where the gain (dB) is 3 dB lower than the maximum gain.

For a transistor to act as an amplifier, it must be properly biased. i.e. its emitter base junction must be forward biased & collector base junction must be reverse biased. The transistor is operating in the active region, is verified by measuring the dc bias conditions.

Circuit diagram for measuring dc bias conditions: -

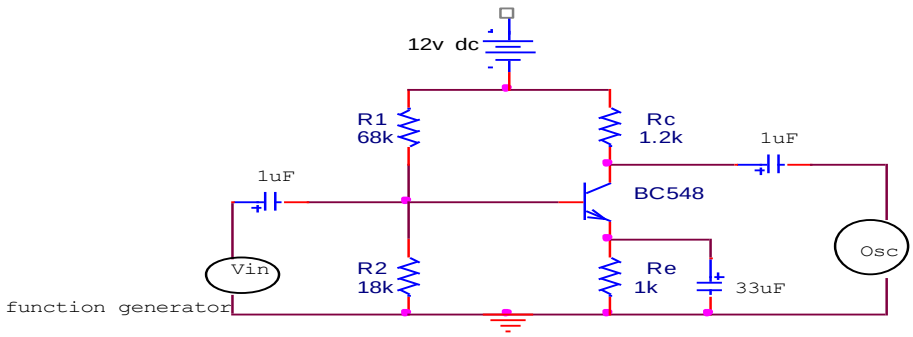
Connect the components as shown in the circuit diagram. Measure the DC bias conditions.

Circuit Diagram: -



Vce = (Approx 25% to 75% of Vcc), Vbe= (Approx. .6V → .7V), Ic= (Approx. VRc /Rc)

Circuit diagram to find the frequency response curve: -



Procedure: -

1. Connect the components as shown in the circuit diagram.
2. Measure the DC bias conditions, before connecting the capacitors.

3. After connecting the capacitors apply a 50mv peak-to-peak signal from the function generator to the input of the amplifier and measure output voltage in the DSO. Note down the Vpk-pk readings for different frequencies keeping the input signal constant.
4. Calculate the voltage gain in dB for each frequency and plot the frequency response curve. From it find the bandwidth.
5. In the mid frequency, select any frequency, go on increasing the input, observing the output in the oscilloscope. At some value of input the output waveform starts showing distortion. That input value is called the signal handling capacity.

Result

1. Measure the DC Bias Voltage

   Vce = (Approx 25% to 75% of Vcc)

   Vbe= (Approx .6V → .7V)

   Ic= (Approx VRc/Rc)

2. AC gain =Vo/Vin

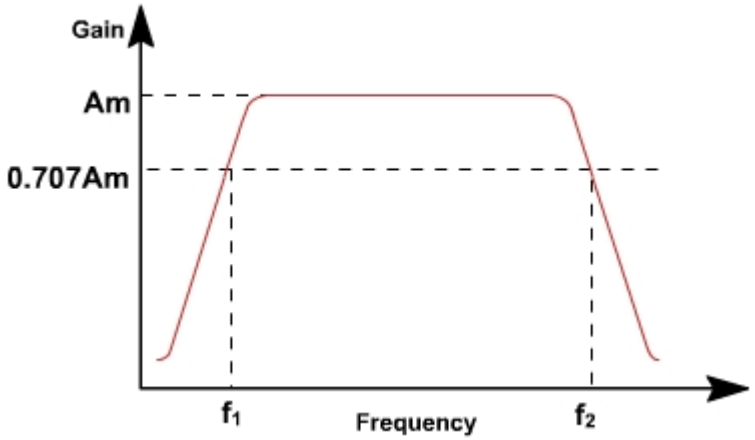
3. Bandwidth measurement(f2-f1)

4. Signal Handling Capacity

Table

V_{in}	Frequency(f)	V_{out}	Gain (V_{out}/V_{in})	Gain in dB (20Log V_{out}/V_{in})
50mVolts p-p	50Hz			
50mVolts p-p	100Hz			
50mVolts p-p	500Hz			
50mVolts p-p	1Khz			
50mVolts p-p	5khz			
50mVolts p-p	10KHz			
50mVolts p-p	20Khz			
50mVolts p-p	50KHz			
50mVolts p-p	100KHz			
50mVolts p-p	200KHz			
50mVolts p-p	500KHz			
50mVolts p-p	700KHz			
50mVolts p-p	1MHz			
50mVolts p-p	2MHz			
50mVolts p-p	3MHz			
50mVolts p-p	4MHz			

Expected Graph- Plot the graph in a semi log graph paper


Gain
Am
0.707Am
f1
Frequency
f2

Experiment 7 Design a Low-Pass Filter and High-Pass Filter

Design a low Pass Filter and High Pass Filter

Objective: - 1) Design the high pass and low pass filter.2) Calculate the lower cut off and higher cut-off Frequency.3) Calculate the gain.

Equipment's Required: - 1) Function Generator 2) Bread Board 3) DSO.

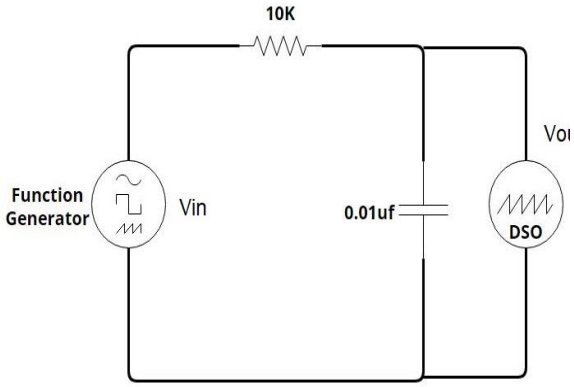
Components required: - Resistors (10K Ohm) Capacitors (0.01uF)

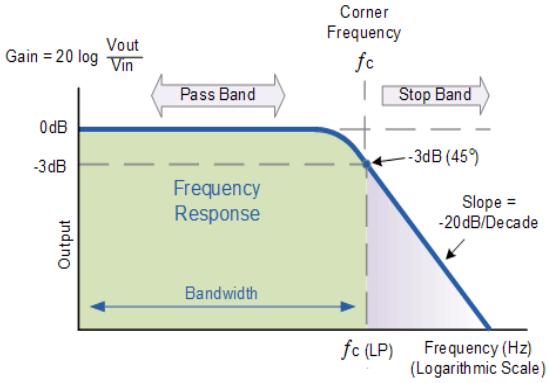
Theory: -A **filter** is a circuit that passes a specific range of frequencies while rejecting other frequencies. A **passive filter** consists of passive circuit elements, such as capacitors, inductors, and resistors. The most common way to describe the frequency response of a filter is to plot the filter voltage gain (V_{out}/V_{in}) in **dB** as a function of frequency **(f).** The frequency at which the output power gain drops to **50%** of the maximum value is called the **cut-off frequency (f_c).** When the filter **dB** voltage gain is plotted as a function of frequency on a semi log graph using straight lines to approximate the actual frequency response, it is called a **Bode plot.** A Bode plot is an ideal plot of filter frequency response because it assumes that the voltage gain remains constant until the cut-off frequency is reached. The filter network voltage gain in **dB** is calculated from the actual voltage gain **(A)** using the equation

$\mathbf{A_{dB}} = 20 \log A$ where $\mathbf{A} = V_{out}/V_{in}$

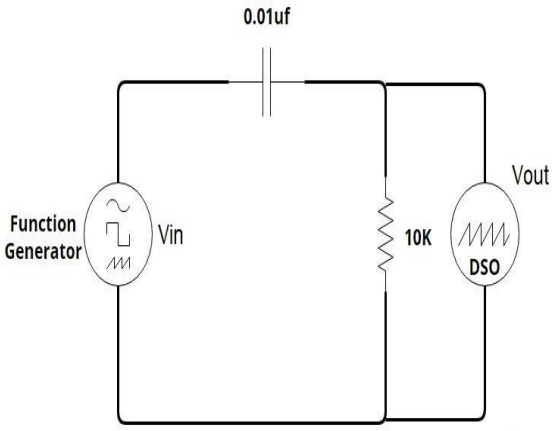
Circuit Diagram

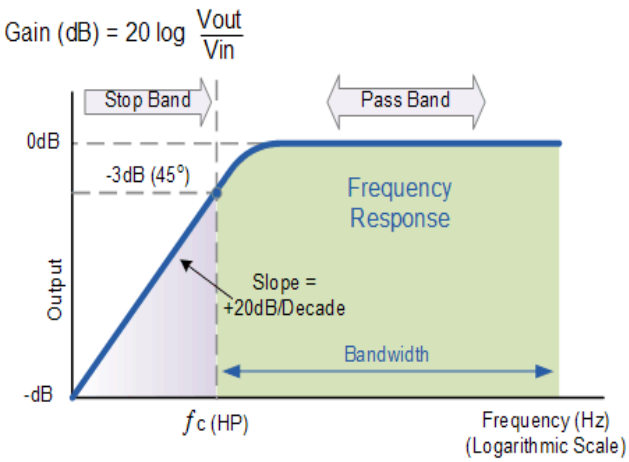
Low pass filter





High Pass Filter





PROCEDURE

1. Set up the circuit as shown taking the output across the capacitor (For HPF set the circuit as shown and take the output across resistor). The input for the filter is taken from output of function generator. The output is connected to DSO.

2. Vary the frequency of the input signal over a wide frequency range (but keep the input amplitude fixed). Note the Values of $V_{out\ for}$ each frequency and calculate the corresponding Gain.

3. Plot the values of Gain vs Frequency in a semi-log graph paper and find out the cut-off frequency from it (higher cut-off for LPF and lower cut-off for HPF).

Bandwidth & cut off frequency measurement of LPF and HPF.

$V_{in=1Volt(pk-pk)}$	Frequency(f_{in})	V_{out}	Gain (V_{out}/V_{in})	Gain in dB ($20Log\ V_{out}/V_{in}$)
	50Hz			
	100Hz			
	500Hz			
	1Khz			
	5khz			
	10KHz			
	20Khz			
	50KHz			
	100KHz			
	200KHz			
	500KHz			
	700KHz			
	1MHz			

Experiment 8 Design of an Inverting and Non-Inverting Amplifier Using Op-Amp IC741

Design an inverting and Non-Inverting Amplifier using OPAMP(IC741)

Objective: -

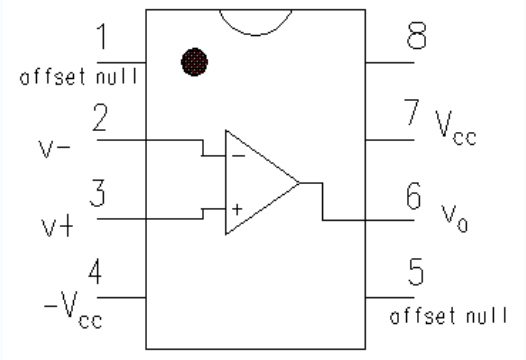
1. Design the circuit diagram. Determine the theoretical output voltage of inverting and non-inverting amplifier. Take the reading and draw the input & output waveform

Equipment's required: - 1) Multimeter, CRO, Bread Board, Function generator for input source, Power Supply ±15.

Components Required: -1) Op Amp (IC741) – 1 No.2) Resistors R_{in}=1k Ohm, Rf= 3.9K Ohm.

Theory: -An Operational Amplifier or OPAMP is a very high gain differential amplifier with high input impedance and low output impedance. The basic circuit is made using a difference amplifier having two inputs and one output. The plus (+) input produces an output that is in phase with the signal applied, whereas an input to the minus (-) input results in an opposite polarity output.

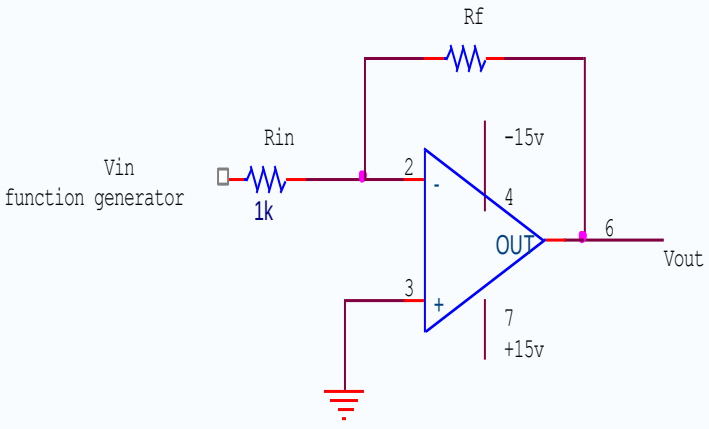
Most existing OPAMPs are produced on a single semi-conductor substrate as an Integrated Circuit (IC741). The Pin diagram is



The two most widely used constant gain amplifier are the inverting and non-inverting amplifier.

Circuit Diagram

Inverting amplifier

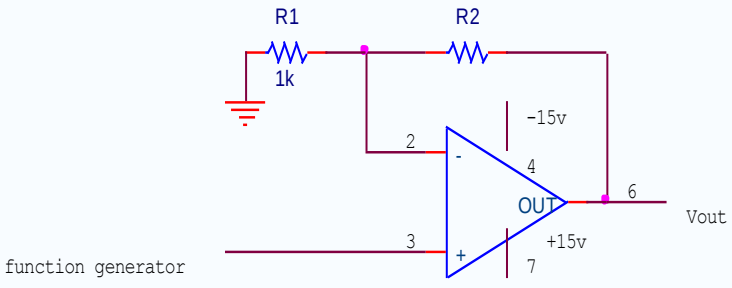


$$V_{out} = -\frac{R_f}{R_{in}} V_{in}$$

Inverts and amplifies a voltage (multiplies by a negative constant)

- $Z_{in} = R_{in}$ (because V_- is a virtual ground)

Non-inverting amplifier



Amplifies a voltage (multiplies by a constant greater than 1)

$$V_{out} = V_{in}\left(1 + \frac{R_2}{R_1}\right)$$

- input impedance $Z_{in} \approx \infty$
  - The input impedance is *at least* the impedance between non-inverting (+) and inverting (−) inputs, which is typically 1 MΩ to 10 TΩ, plus the impedance of the path from the inverting (−) input to ground (i.e., R_1 in parallel with R_2).
  - Because negative feedback ensures that the non-inverting and inverting inputs match, the input impedance is actually **much higher.**

Result Table: -

Type of Amplifier	Input AC voltage 1 volt ,1 khz	Output Voltage	Calculate the theoretical Gain	Measure Practical gain	Measure phase difference
1. The Inverting OPAMP					
Non-Inverting					

PROCEDURE: -

- The OPAMP was set up as per the circuit diagram.
- Input voltage of 1 volt,1 khz was provided and the gain of the amplifier was measured by noting down the output voltage.
- The gain was determined with feedback resistor (R_f) keeping the input resistance constant (R_{in}).

Experiment 9 Study of Diode as Clipper and Clamper

Study of diode as a Clipper & clamper Circuit

Objective: - Study of the diode as a clipper & clamper. Draw and measure the input and output waveform.

Required Components: - Resistor 1K -1 No, Diode (1N4007) -1No, A.C Voltage, DC Voltage, **DSO,** Coupling wires, Multi meter

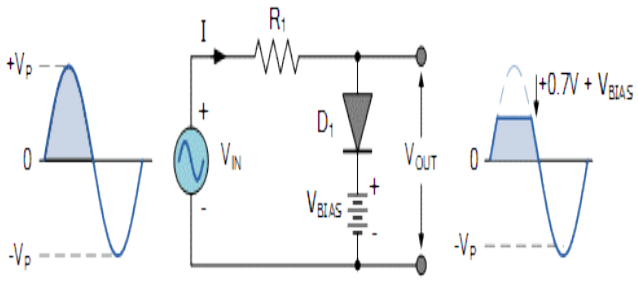
Theory: -

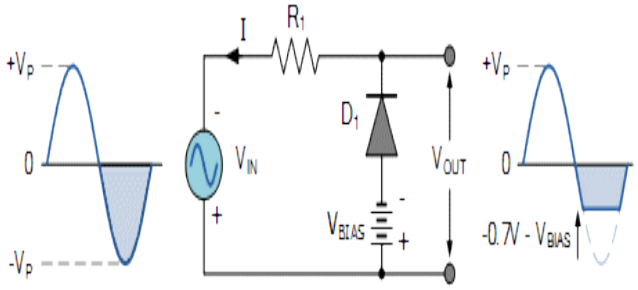
For a clipping circuit at least two components—an ideal diode and resistor are required and sometimes a dc battery is also employed for fixing the clipping level. The diode acts as a closed switch when forward biased and an open switch when reverse biased. Depending on the orientation of the diode, the positive or negative region of the input signal is "clipped" off and accordingly the diode clippers may be positive or negative clippers. Half wave rectifier circuits can also be called the basic clippers.

Biased Clipper – The level to which an ac voltage is limited can be adjusted by adding a bias voltage V_{BIAS}, in series with the diode. Biased clippers are employed for this purpose. The circuit diagram for a biased positive clipper (that is for removing a small portion of positive half cycle) is illustrated in figure. When the input signal voltage is positive but does not exceed DC voltage V_{BIAS}, the diode D remains reverse biased and most of the input voltage appears across the output. When during the positive half cycle of input signal, the signal voltage exceeds the DC voltage V_{BIAS}, the diode D is forward biased & conducts heavily.

Procedure: - 1) Connect the components according to the circuit diagram given. Take V_{IN}=5V rms, V_{BIAS}=2.5V 2) Connect DSO across the load R and input source. 3) Simultaneously measure both the input and output waveform in DSO ch-1 and ch-2 respectively.

Positive Bias Diode Clipping **Negative Bias Diode Clipping**

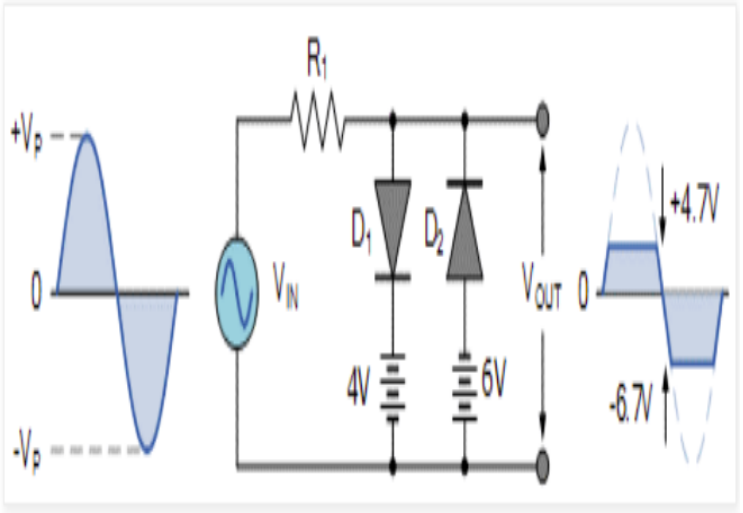




- **Likewise, by reversing the diode and the battery bias voltage, when a diode conducts the negative half cycle of the output waveform is held to a level $-V_{BIAS} - 0.7V$ as shown.**

Double sided Diode Clipping

A variable diode clipping or diode limiting level can be achieved by varying the bias voltage of the diodes. If both the positive and the negative half cycles are to be clipped, then two biased clipping diodes are used. But for both positive and negative diode clipping, the bias voltage need not be the same. The positive bias voltage could be at one level, for example 4 volts, and the negative bias voltage at another, for example 6 volts as shown.



Diode Clipping at Different Bias levels
***take Vin = 7v Rms ***

When the voltage of the positive half cycle reaches +4.7 V, diode D_1 conducts and limits the waveform at +4.7 V. Diode D_2 does not conduct until the voltage reaches –6.7 V. Therefore, all positive voltages above +4.7 V and negative voltages below –6.7 V are automatically clipped.
The advantage of biased diode clipping circuits is that it prevents the output signal from exceeding preset voltage limits for both half cycles of the input waveform, which could be an input from a noisy sensor or the positive and negative supply rails of a power supply.
If the diode clipping levels are set too low or the input waveform is too great then the elimination of both waveform peaks could end up with a square-wave shaped waveform.

DIODE as a Clamper

Object: - Study of diode as a positive clamper

Required Components: -Capacitor C=1uF 1No, Resistor R=680K Ohm, Diode (1N4007)-1No, A.C Voltage Source, DC Voltage, DSO

Theory: -A Clamper Circuit is a circuit that adds a DC level to an AC signal. A Clamper circuit can be defined as the circuit that consists of a diode, a resistor and a capacitor that shifts the waveform to a desired DC level without changing the actual appearance of the applied signal. The different types of clampers are positive and negative clampers.

The magnitude of R and C must be chosen such that the time constant RC is large enough to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is non- conducting. A Clamping circuit restores the DC level. When a negative peak of the signal is raised above to the zero level, then the signal is said to be **positively clamped**. A Positive Clamper circuit is one that consists of a diode, a resistor and a capacitor and that shifts the output signal to the positive portion of the input signal. The figure below explains the construction of a positive clamper circuit.

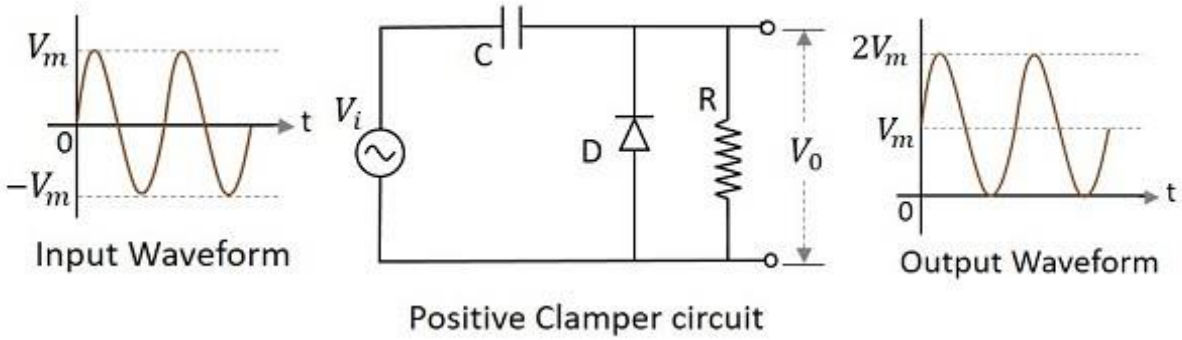
Initially when the input is given, the capacitor is not yet charged and the diode is reverse biased. The output is not considered at this point of time. During the negative half cycle, at the peak value, the capacitor gets charged with negative on one plate and positive on the other. The capacitor is now charged to its peak value V_m. The diode is forward biased and conducts heavily. During the next positive half cycle, the capacitor is charged to positive Vm while the diode gets reverse biased and gets open circuited. The output of the circuit at this moment will be

$$V_0 = V_i + V_m$$

Hence the signal is positively clamped as shown in the above figure. The output signal changes according to the changes in the input, but shifts the level according to the charge on the capacitor, as it adds the input voltage.

Positive Clamper

The circuit for a positive clamper is shown in the figure. During the negative half cycle of the input signal, the diode conducts and acts like a short circuit. The output voltage V_0 =0V.The capacitor is charged to the peak value of input voltage V_m. and it behaves like a battery. During the positive half of the input signal, the diode does not conduct and acts as an open circuit. Hence the output voltage V0 =Vm+Vm. This gives a positively clamped voltage**. To see the output in DSO keep it in DC coupling mode**.



Experiment 10:

Verification of Basic Logic Gates

Verification of basic logic gates

Verification and interpretation of truth tables for AND, OR, NOT, NAND, NOR Exclusive OR (EX-OR) Gates. Implement AND, OR, NOT functions using universal gates NAND & NOR

Apparatus: Universal Work Station, logic gates ICs, wires,2mm banana connectors

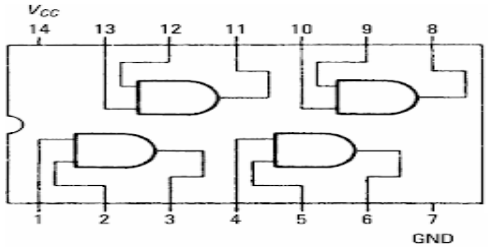
Theory: Logic gates are electronic circuits which perform logical functions on one or more inputs to produce one output. When all the input combinations of a logic gate are written in a series and their corresponding outputs written along them, then this input/ output combination is called **Truth Table**.

Procedure:

1.Connect the V_{CC} +5v to the IC pin 14 w.r.t Ground pin 7.
2.Connect the inputs of logic gate to the logic switches and its output to the logic indicator (LED).
3.Apply various input combinations and observe output for each one.
4.Verify the truth table for each input/ output combination.
5.Repeat the process for all other logic gates.

AND Gate (7408)

AND gate produces an output as 1, when all its inputs are 1; otherwise, the output is 0. This gate can have minimum 2 inputs but output is always one. Its output is 0 when any input is 0.

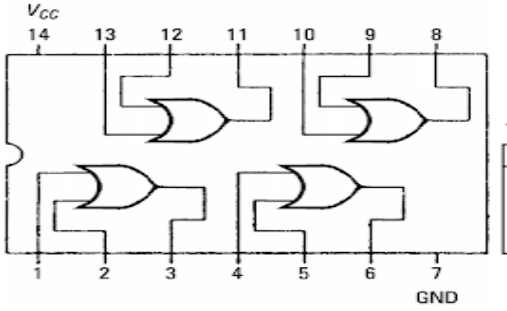


Y=A.B

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate (7432)

OR gate produces an output as 1, when any or all its inputs are 1; otherwise, the output is 0. This gate can have minimum 2 inputs but output is always one. Its output is 0 when all input is 0.

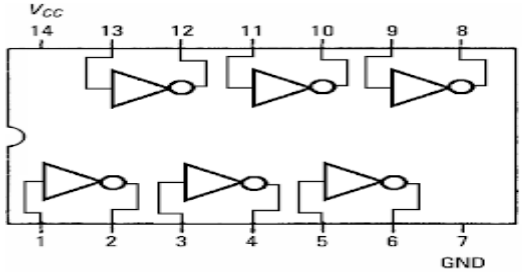


Y=A+B

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT Gate (7404)

NOT gate produces the complement of its input. This gate is also called an INVERTER. It always has one input and one output. Its output is 0 when input is 1 and output is 1 when inputs are 0.

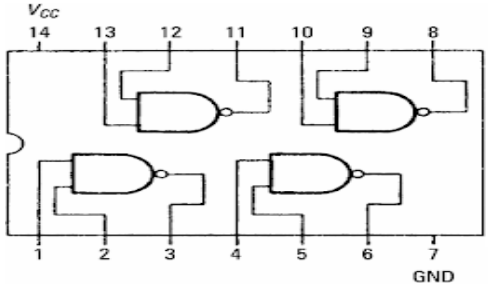


$Y=\overline{A}$

A	Y
0	1
1	0

NAND Gate (7400)

NAND gate is actually a series of AND gate with NOT gate. If we connect the output of an AND gate to the input of a NOT gate, this combination will work as NOT-AND or NAND gate. Its output is 1 when any or all inputs are 0, otherwise output is 1.

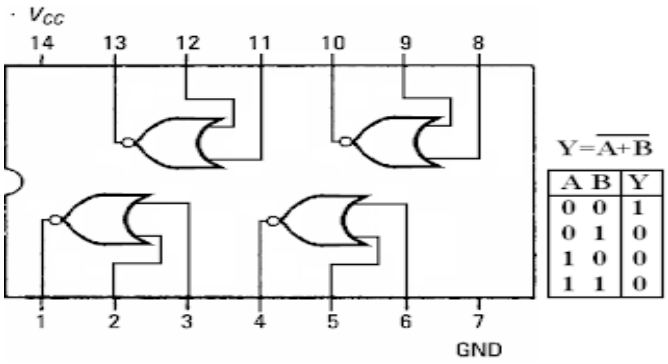


$Y=\overline{A.B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate (7402)

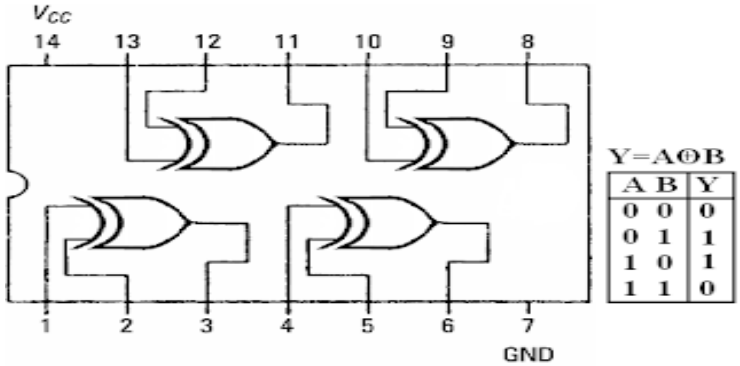
NOR gate is actually a series of OR gate with NOT gate. If we connect the output of an OR gate to the input of a NOT gate, this combination will work as NOT-OR or NOR gate. Its output is 0 when any or all inputs are 1, otherwise output is 1.



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Exclusive OR (X-OR) Gate (7486)

X-OR gate produces an output as 1, when number of 1's at its inputs is **odd**, otherwise output is 0. It has two inputs and one output.

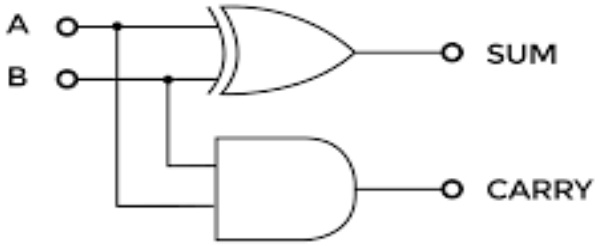


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

- Implement All BASIC Gates using the universal gate **(NAND & NOR).**

Adder and Subtractor

Half adder



A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

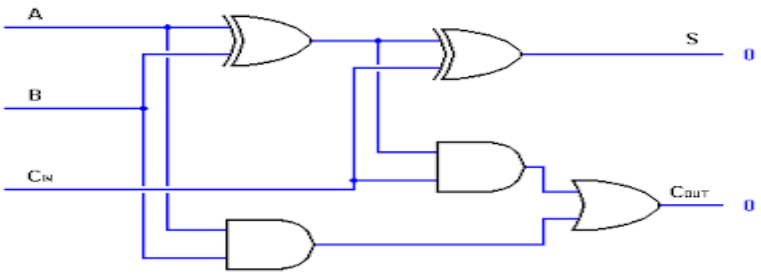
Here the output '1'of '10' becomes the carry-out. The result is shown in a truth-table below. 'SUM' is the normal output and 'CARRY' is the carry-out. From the equation it is clear that this 1-bit adder can be easily implemented with the help of XOR Gate for the output 'SUM' and an AND Gate for the carry. For complex addition, there may be cases when you have to add two 8-bit bytes together. This can be done only with the help of full-adder logic.

Full Adder: - This type of adder is a little more difficult to implement than a half-adder. The main difference between a half-adder and a full-adder is that the full-adder has three inputs and two outputs. The first two inputs are A and B and the third input is an input carry designated as CIN. The output carry is designated as COUT and the normal output is designated as S. Take a look at the truth-table.

Truth Table

INPUTS		OUTPUTS		
A	B	CIN	COUT	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Logic Diagram



From the above truth-table, the full adder logic can be implemented. We can see that the output S is an EXOR between the input A and the half-adder SUM output with B and CIN inputs. We must also note that the COUT will only be true if any of the two inputs out of the three are HIGH.

The sum (S) of the full-adder is the XOR of A, B, and Cin. Therefore,

Sum, $S=A \oplus B \oplus C_{in}=A'B'C_{in}+A'BC'_{in}+AB'C'_{in}+ABC_{in}$

The carry (Cout) of the half-adder is the AND of A and B. Therefore,

Carry, $Cout = AB+AC_{in}+BC_{in}$

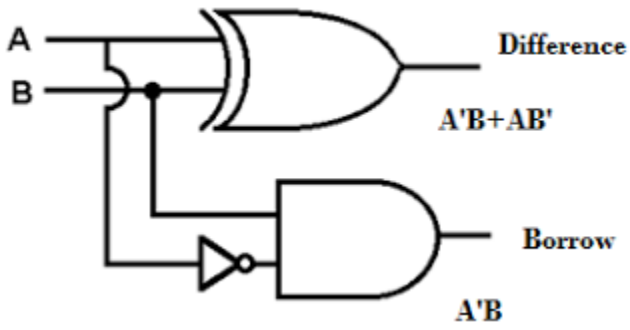
Subtractor: Subtractor is the one which used to subtract two binary number(digit) and provides Difference and Borrow as an output. In digital electronics we have two types of subtractor.

1.**Half Subtractor**

2.Full Subtractor

1.Half Subtractor: Half Subtractor is used for subtracting one single bit binary digit from another single bit binary digit. The truth table of Half Subtractor is shown below.

Logic Diagram

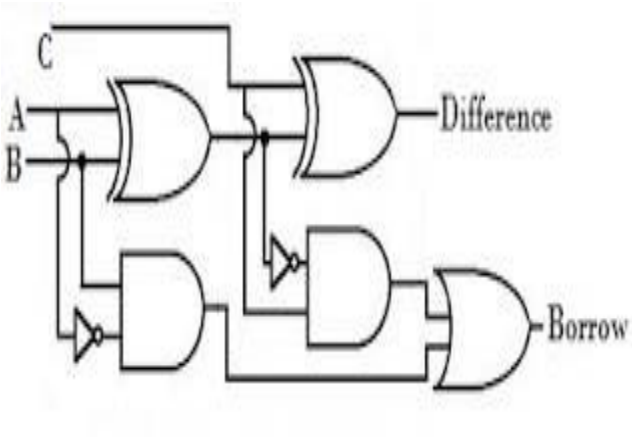


Half Subtractor-Truth Table			
Input		Output	
A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Like Adders Here also we need to calculate the equation of Difference and Borrow
Difference=A'B+AB'=A⊕B
Borrow=A'B

2. Full Subtractor: A logic Circuit Which is used for Subtracting Three Single Bit Binary digit is known as Full Subtractor. The **Truth Table** of Full Subtractor is Shown Below.

LOGIC DIAGRAM



Full Subtractor-Truth Table				
Input			Output	
A	B	C	Difference	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

From_the_Truth_Table_the_Difference_and_Borrow_will_written_as
Difference=A'B'C+A'BB'+AB'C'+ABC=A⊕B⊕C
Reduce_it_like_adder_Then_We_got
Difference=A⊕B⊕C
Borrow=A'B'C+A'BC'+A'BC+ABC
=A'B'C+A'BC'+A'BC+A'BC+A'BC+ABC---------->A'BC=A'BC+A'BC+A'BC
=A'C(B'+B)+A'B(C'+C)+BC(A'+A)
Borrow=A'C+A'B+BC